

A: Datasheet

Algorithm: vtcc_001

Developer: Viettel Cyberspace Center

Submission Date: 2023_12_18

Template size: 4861 bytes

Template time (2.5 percentile): 709 msec

Template time (median): 711 msec

Template time (97.5 percentile): 727 msec

Investigation:

Frontal mugshot ranking 168 (out of 470) -- FNIR(1600000, 0, 1) = 0.0020 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 138 (out of 432) -- FNIR(1600000, 0, 1) = 0.0112 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 62 (out of 401) -- FNIR(1600000, 0, 1) = 0.0690 vs. lowest 0.0510 from nec_009

Immigration visa-border ranking 247 (out of 359) -- FNIR(1600000, 0, 1) = 0.0142 vs. lowest 0.0006 from cloudwalk_mt_002

Immigration visa-kiosk ranking 134 (out of 303) -- FNIR(1600000, 0, 1) = 0.0794 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 27 (out of 470) -- FNIR(1600000, T, L+1) = 0.0021, FPIR=0.001000 vs. lowest 0.0011 from idemia_010

Mugshot webcam ranking 15 (out of 430) -- FNIR(1600000, T, L+1) = 0.0115, FPIR=0.001000 vs. lowest 0.0071 from nec_009

Mugshot profile ranking 3 (out of 400) -- FNIR(1600000, T, L+1) = 0.0791, FPIR=0.001000 vs. lowest 0.0634 from cloudwalk_mt_002

Immigration visa-border ranking 112 (out of 358) -- FNIR(1600000, T, L+1) = 0.0144, FPIR=0.001000 vs. lowest 0.0010 from cloudwalk_mt_002

Immigration visa-kiosk ranking 42 (out of 303) -- FNIR(1600000, T, L+1) = 0.0894, FPIR=0.001000 vs. lowest 0.0485 from nec_009

B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (vtcc_001)

False negative identification rate, FNIR(N)

1e+06

3e+06

1e+07

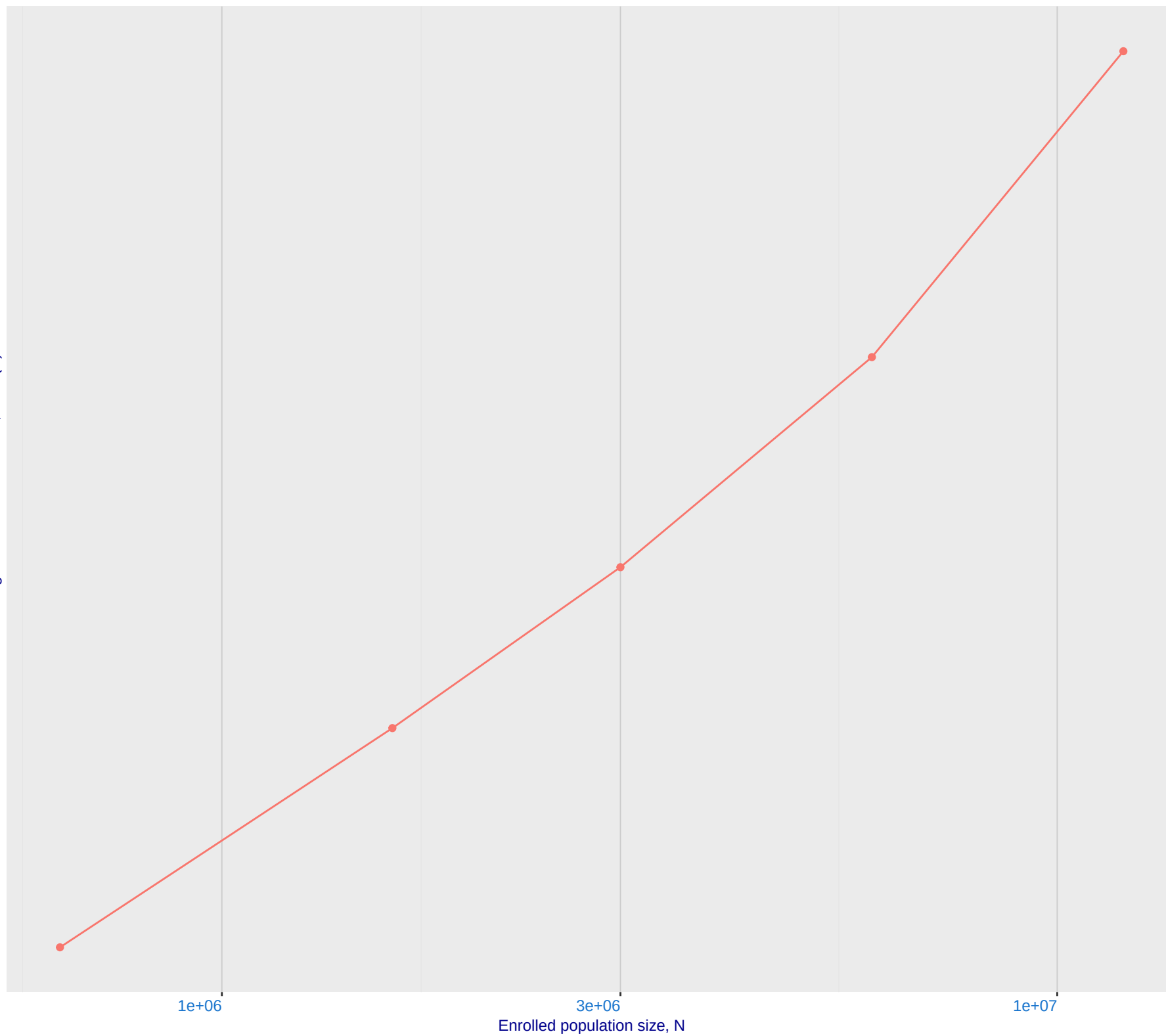
Enrolled population size, N

enrolment_style

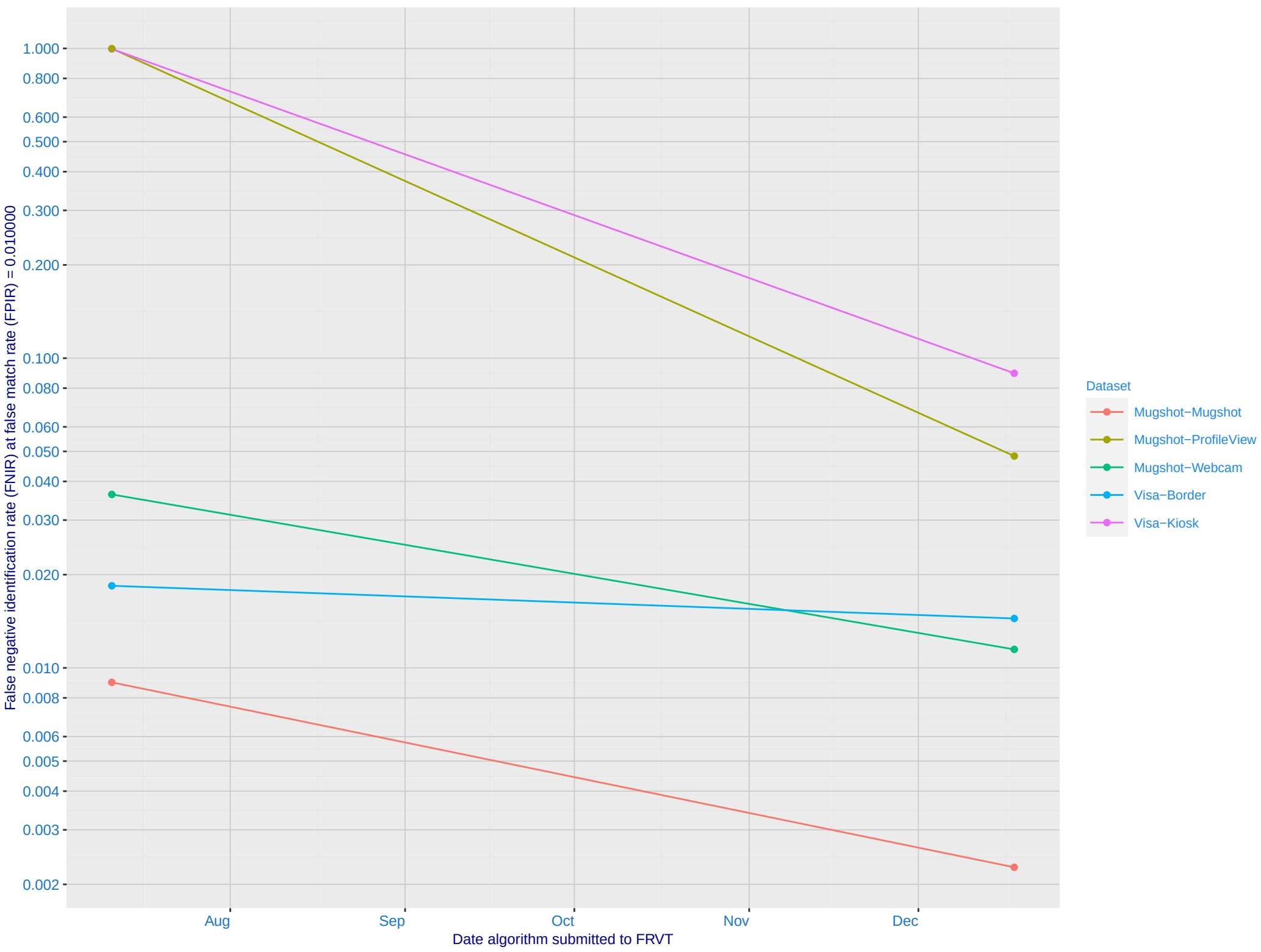
recent

Dataset:
2018 Mugshot
FNIR@FPIR = 0.001

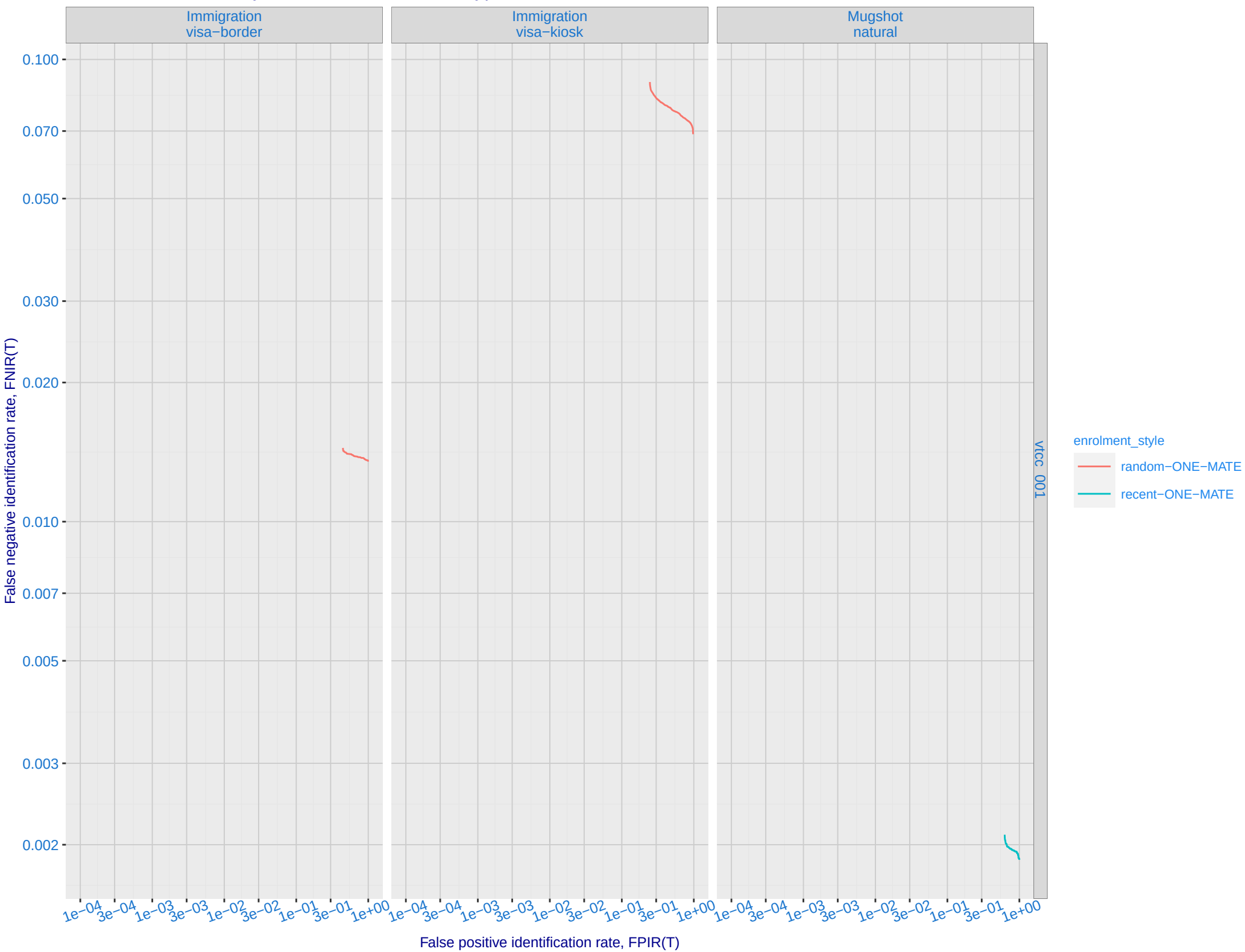
vtcc_001



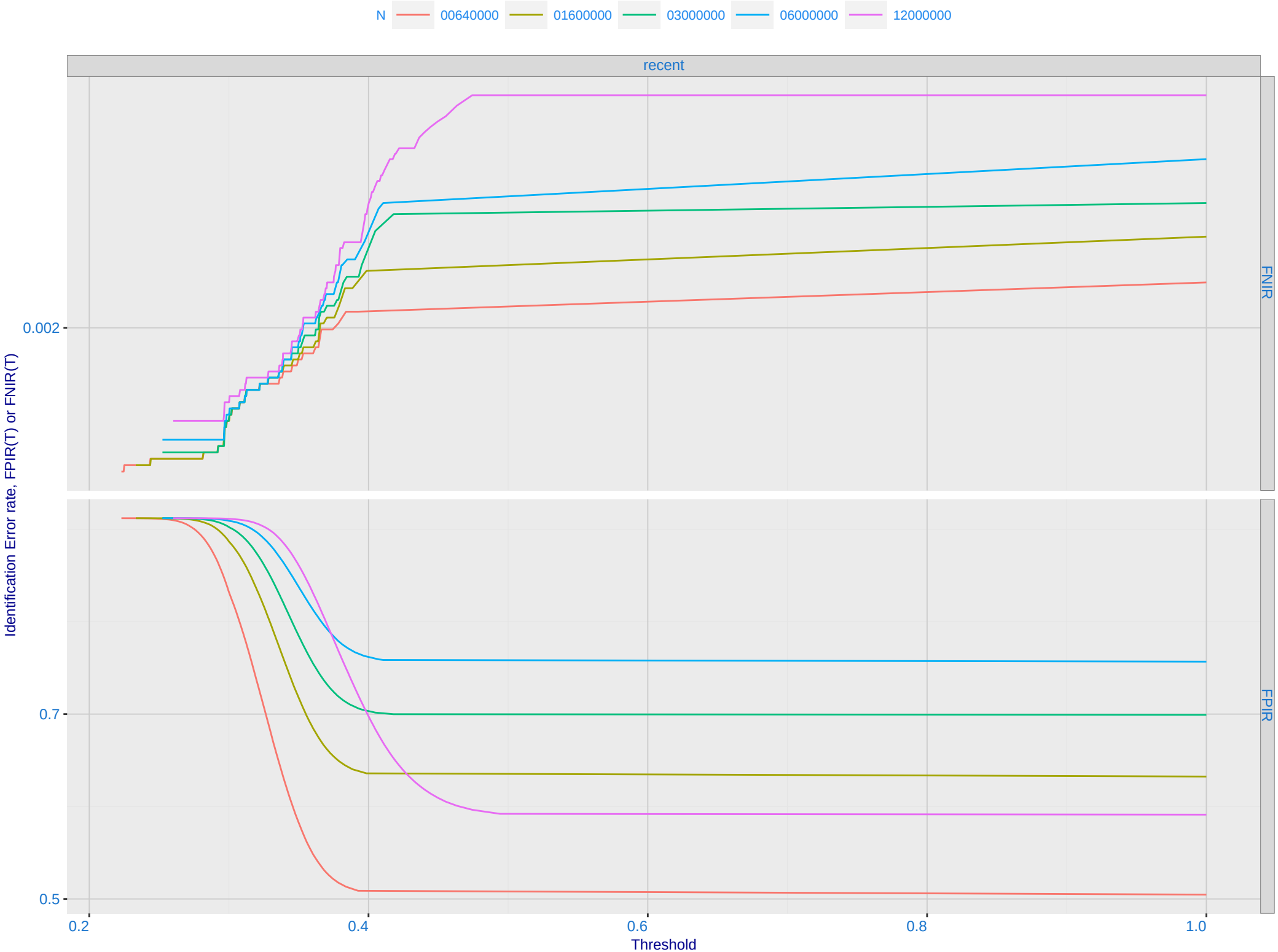
C: Evolution of accuracy for VTCC algorithms on three datasets 2018 – present



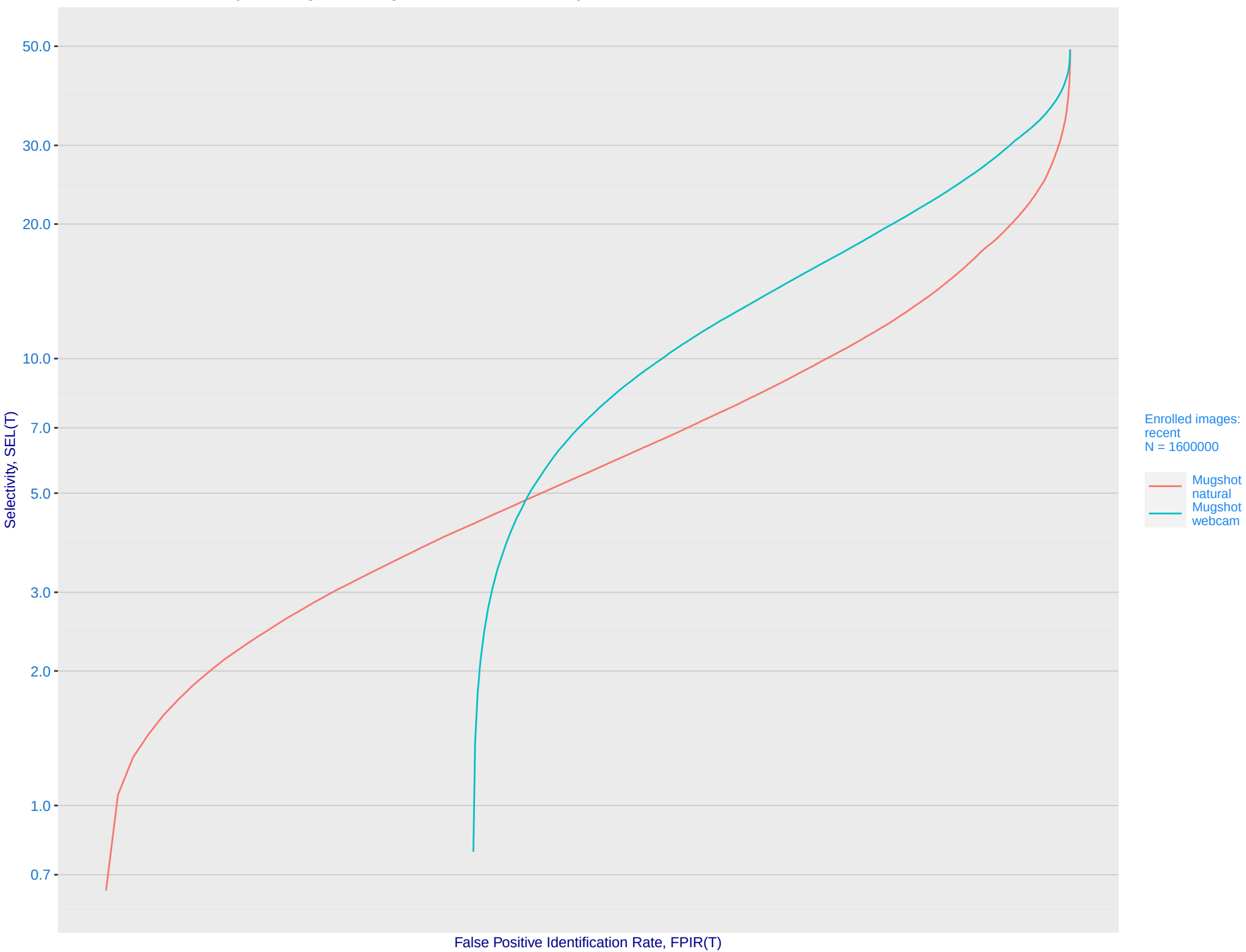
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals



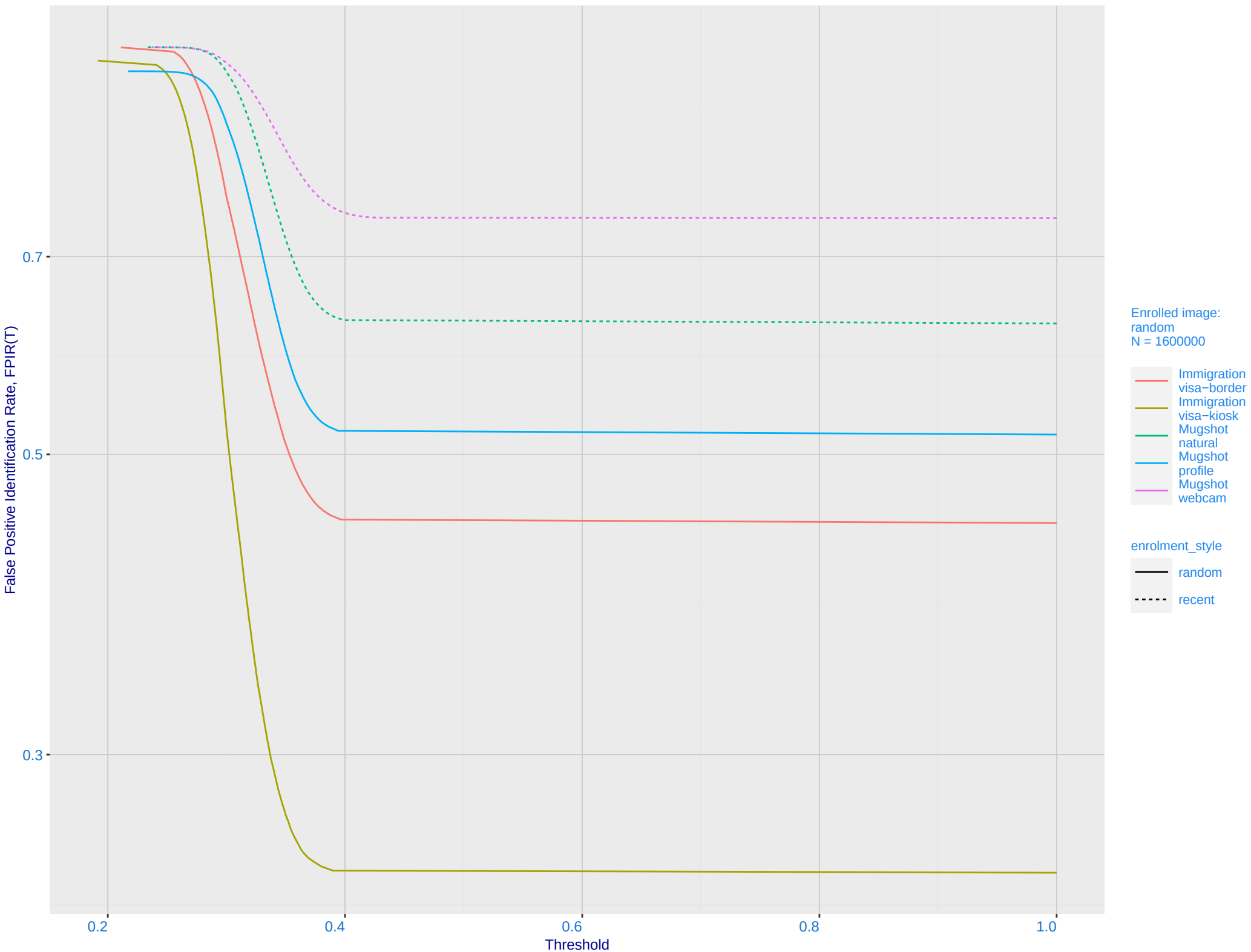
E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images



F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate

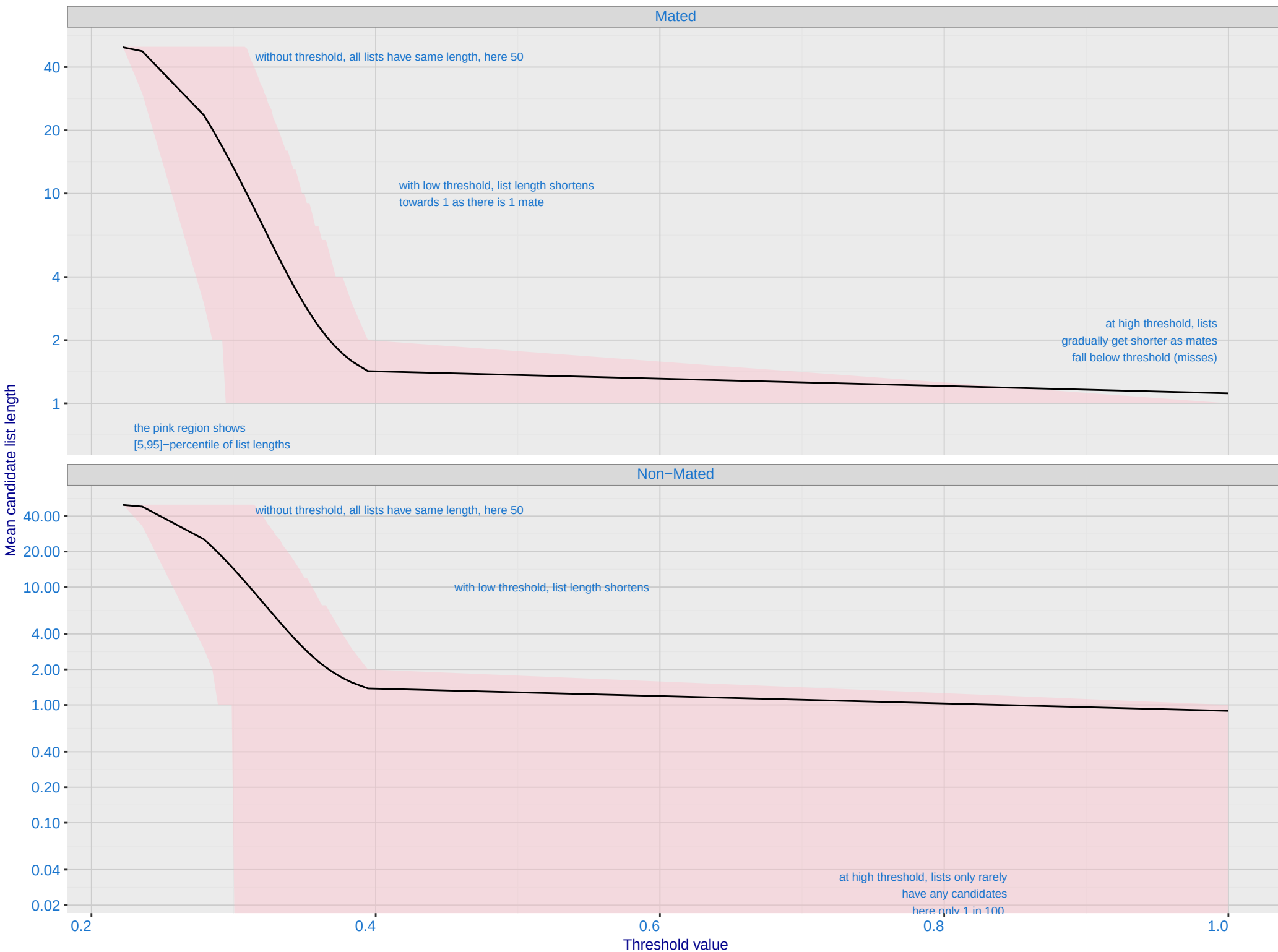


G: FPIR dependence on T by probe type for N = 1600000 subjects



H: Reduced length candidate lists for human review

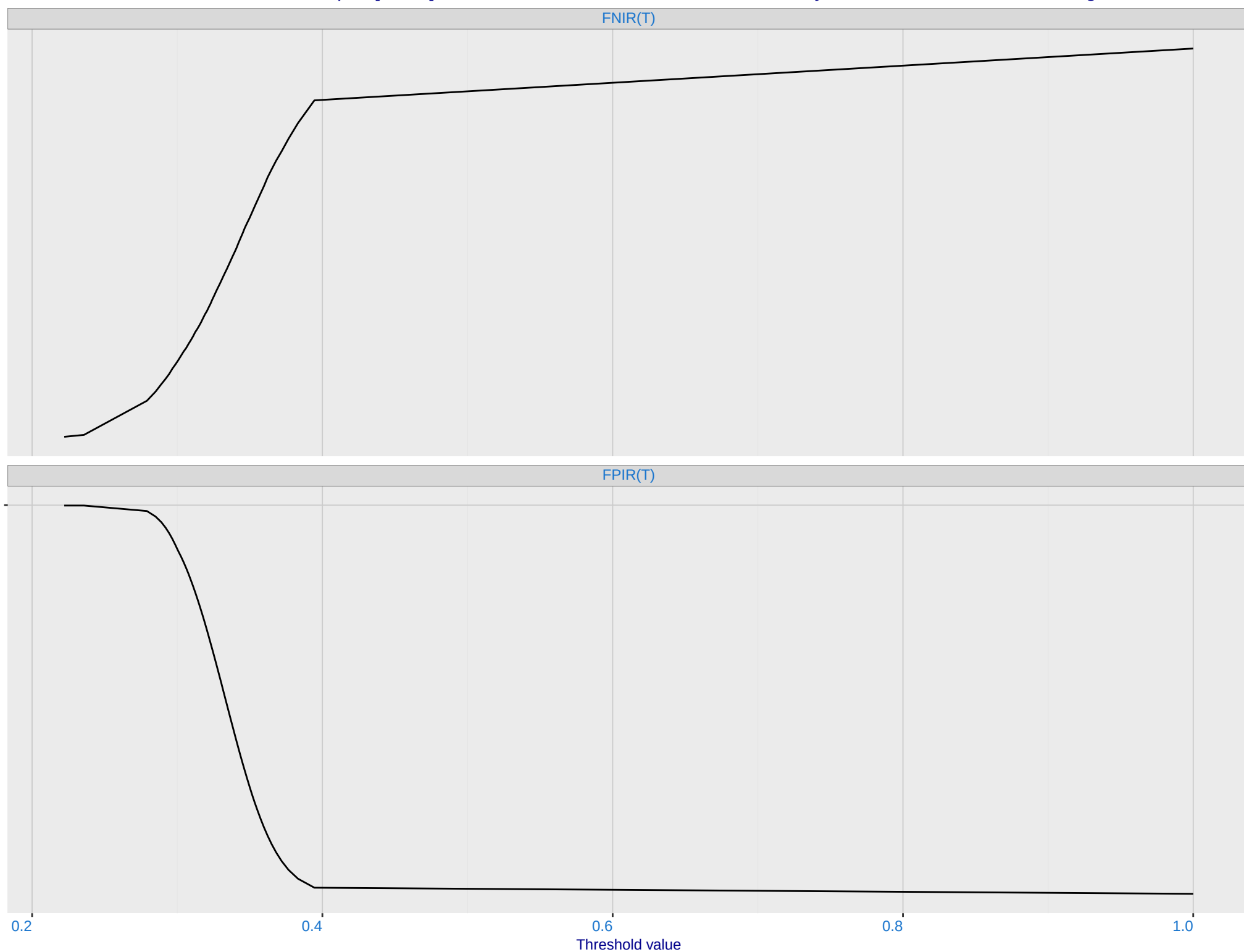
Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image



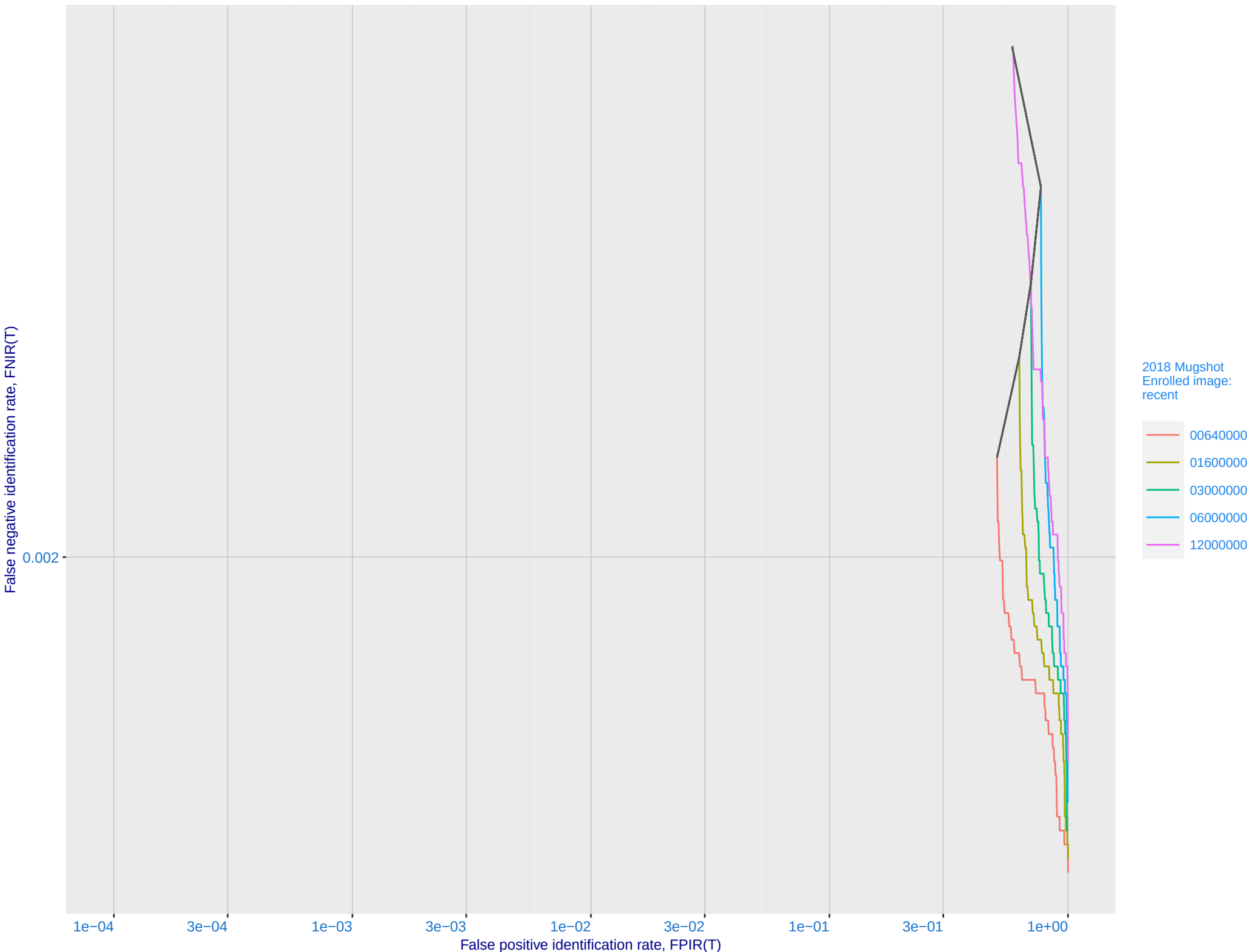
I: FNIR and FPIR dependence on threshold

Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image

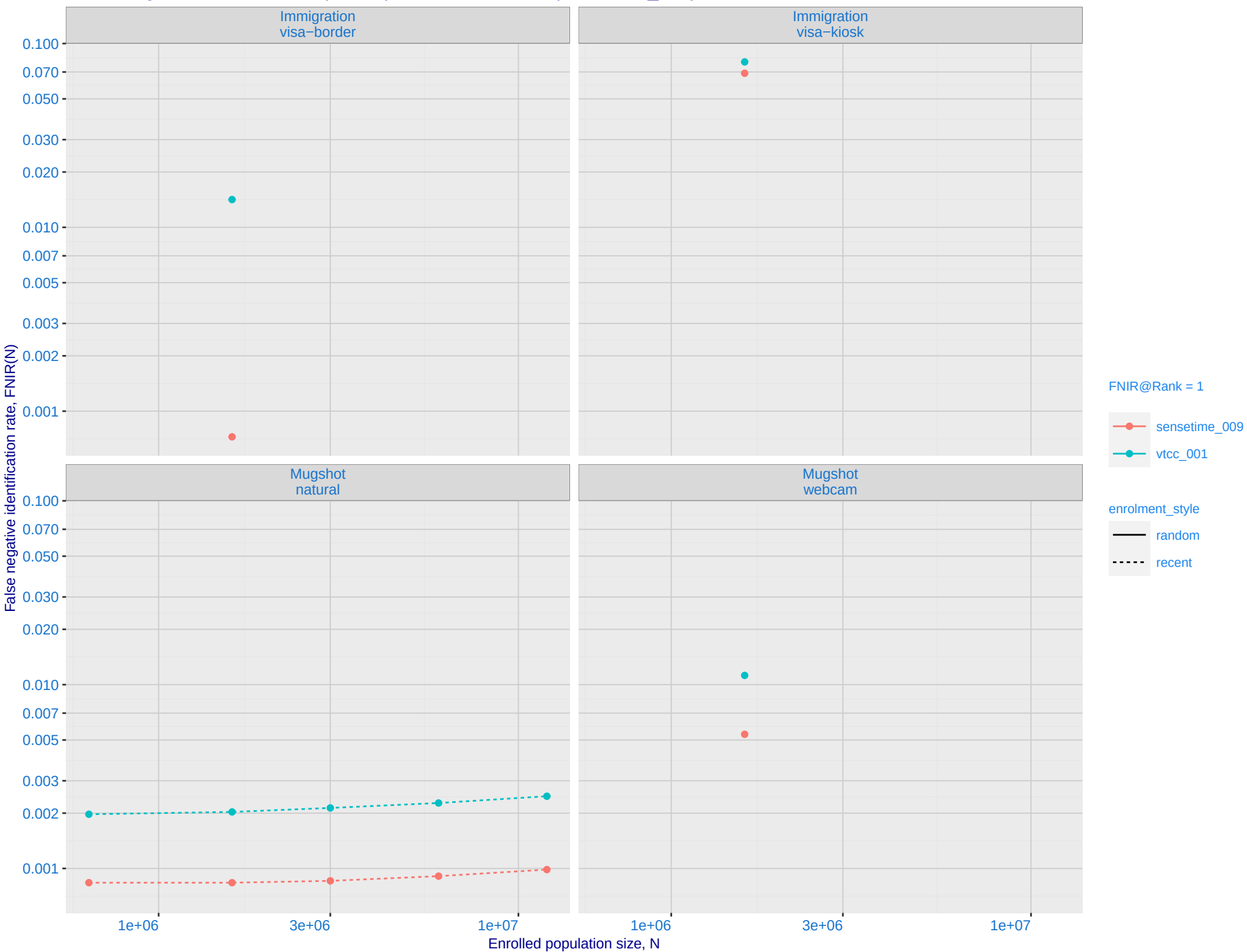
False {Negative | Positive} Identification Rate



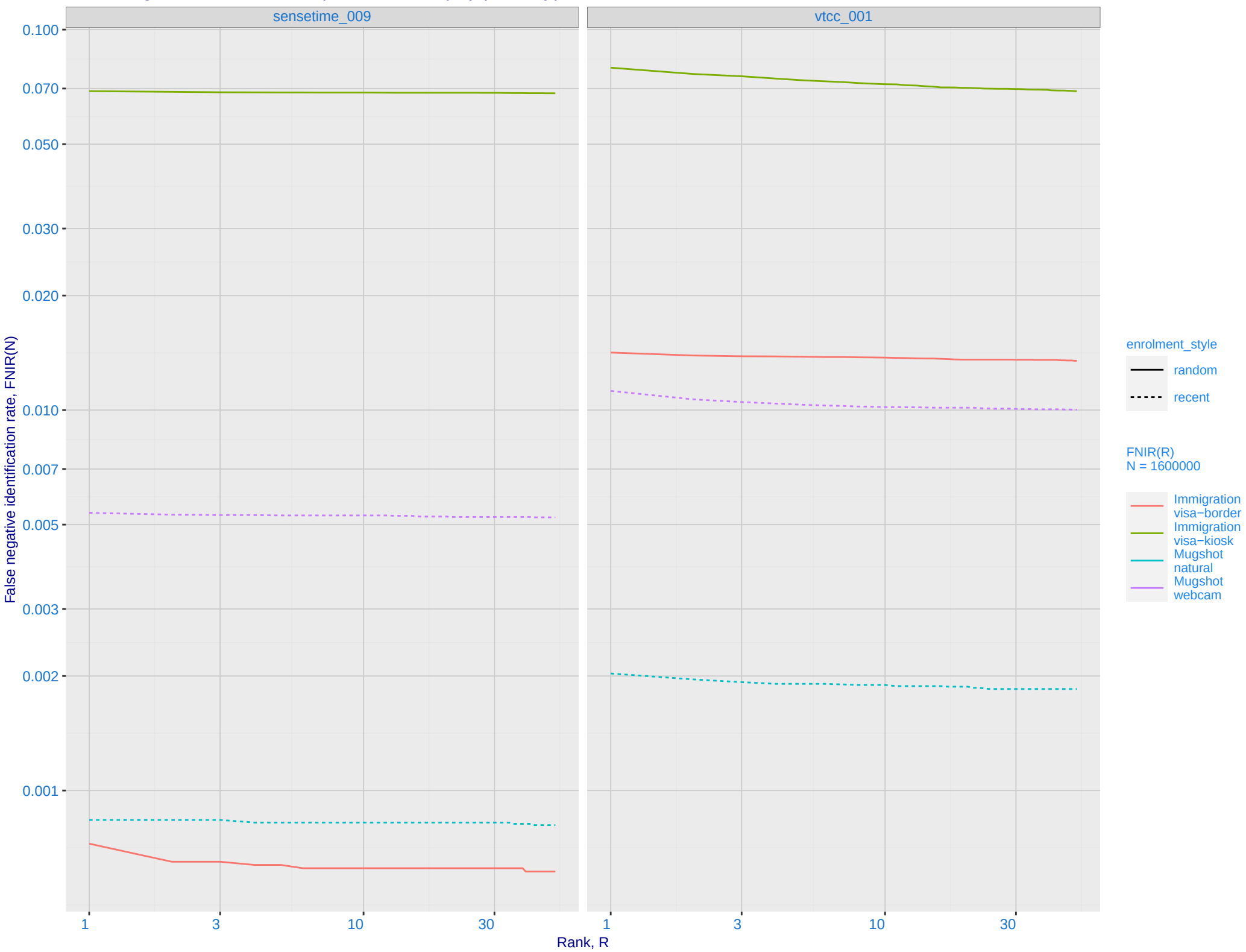
J: DET for Mugshot natural images and various N. Links connect points of equal threshold.



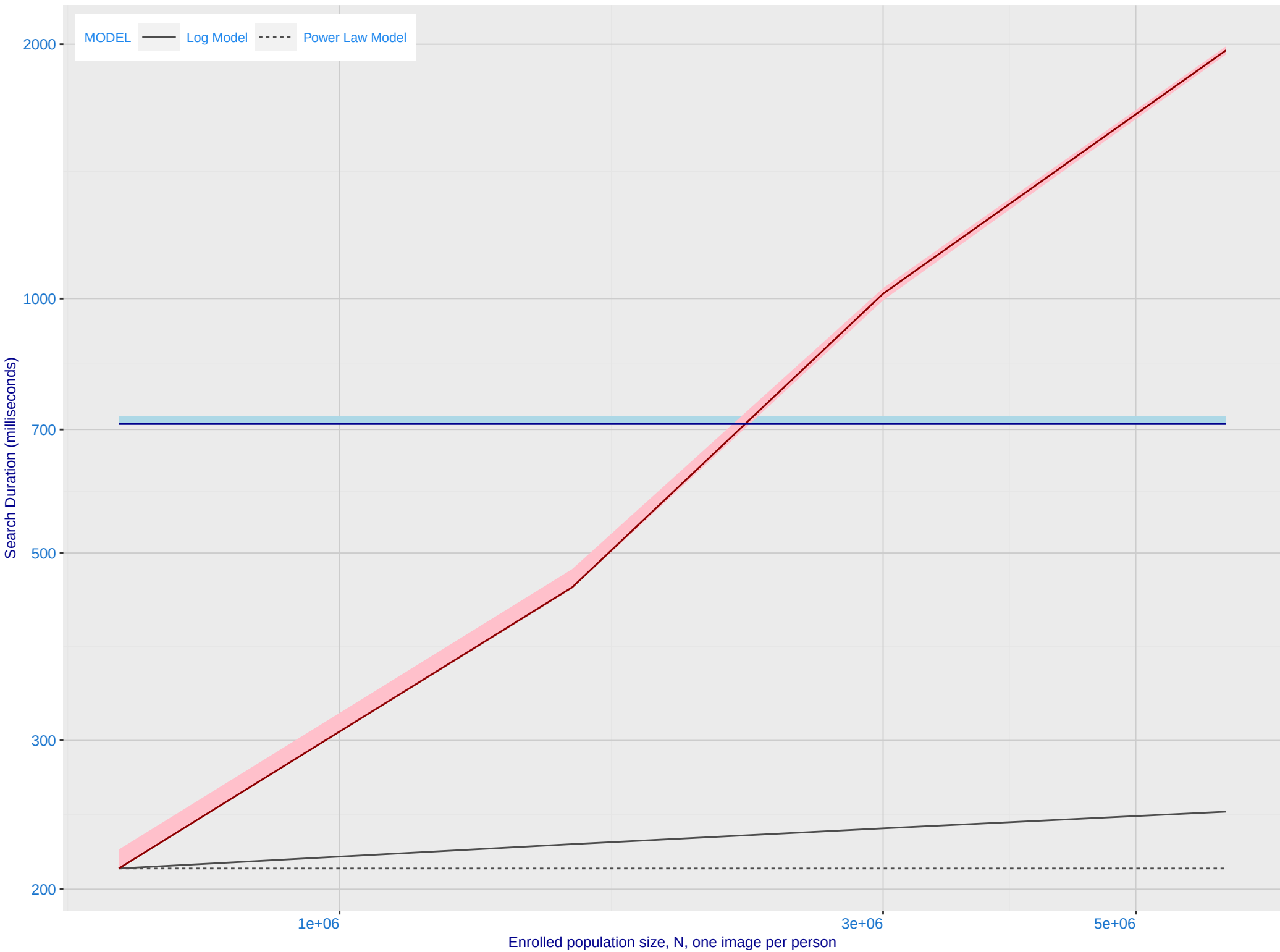
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)



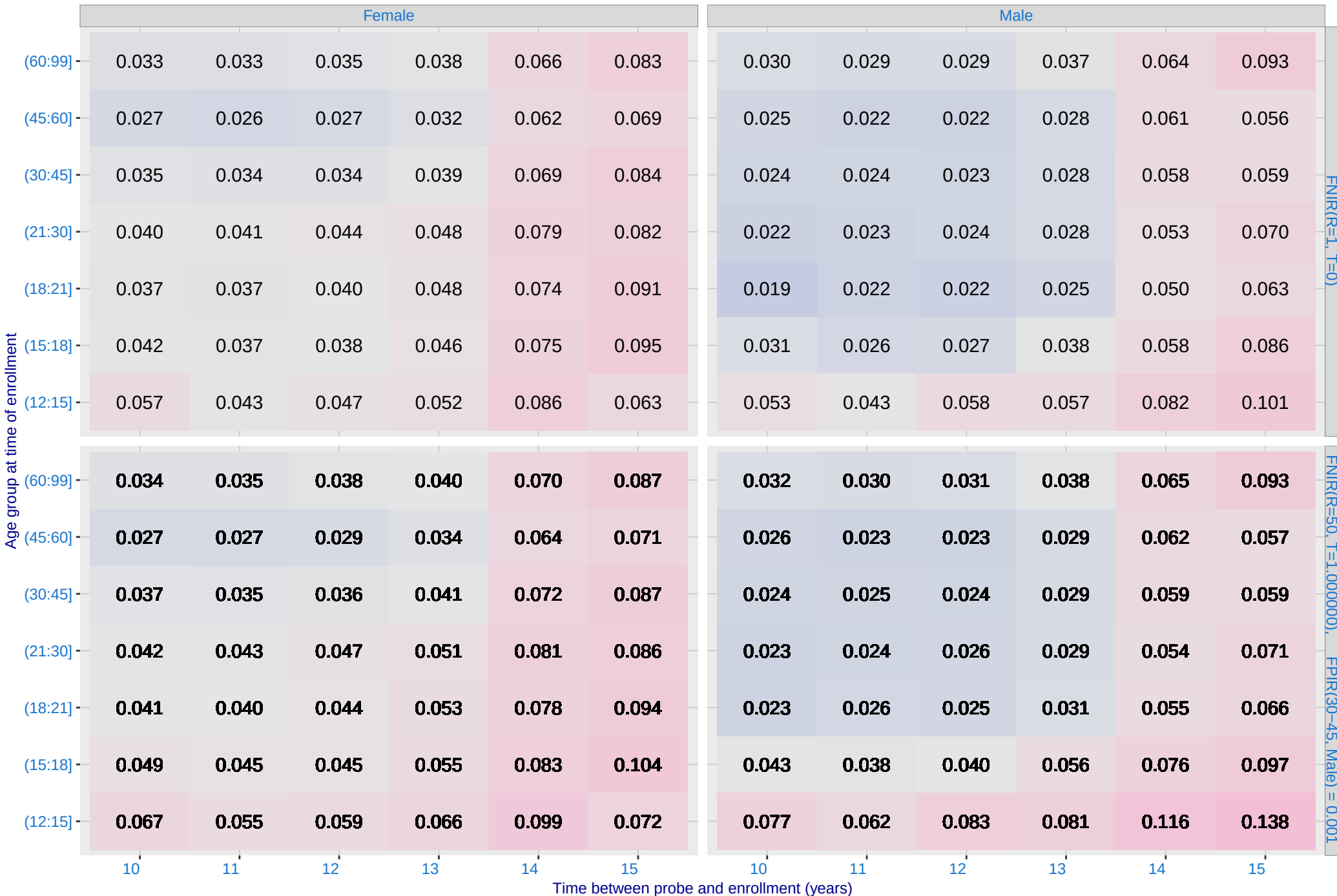
L: Investigational mode: FNIR(1600000, R, 0) by probe type



M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements. The template generation time is independent of N. The log and power-law models are fit to the first two (N,T) observations



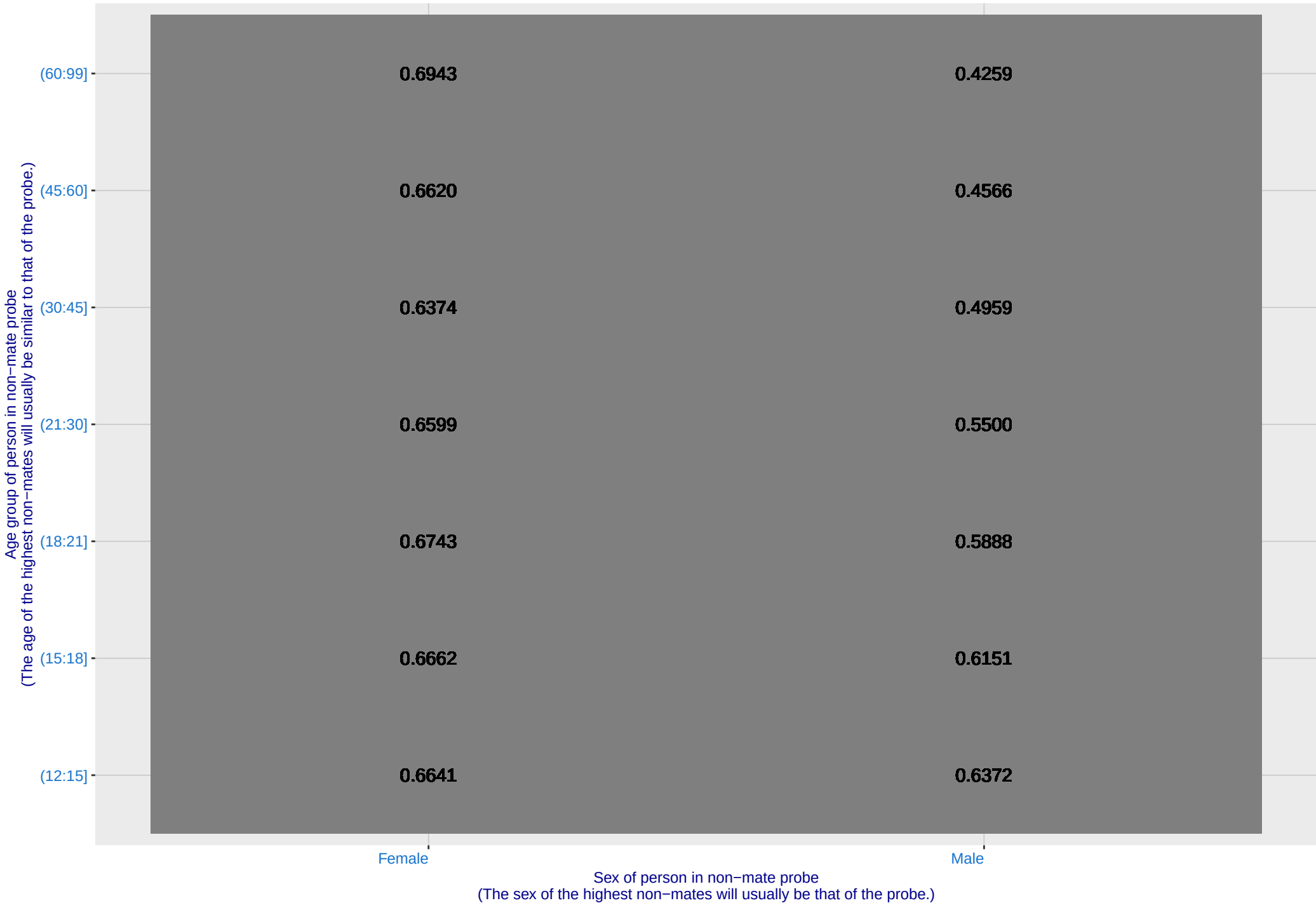
O: FNIR(T, N = 1.6 million) by sex, age and time-lapse. The top row gives investigational rank-1 miss rates. The bottom panels give high threshold for more lights-out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.

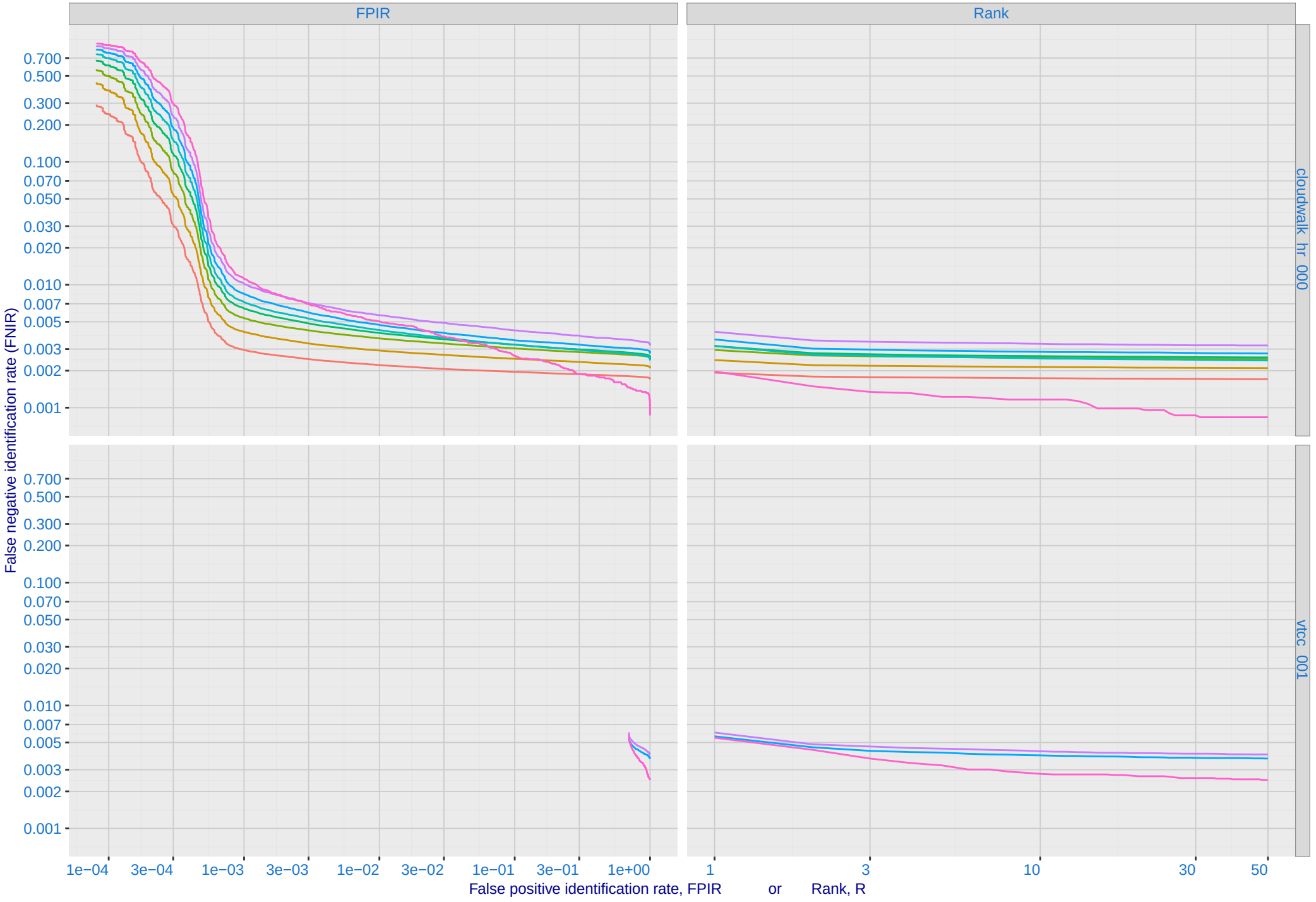
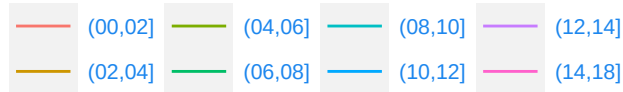
Algorithm: vtcc_001, Dataset: Border-Crossing Ageing
Threshold: 1.000000 set to achive FPIR(30-45, Male) = 0.001

Color encodes log(FPIR)



Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing

Dataset: 2018 Mugshot N = 3068801



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines

